

Multivariate Newton Methods

Learning objectives

After studying this chapter, you should be able to

- derive the multivariate Newton equation from the quadratic Taylor model;
- distinguish model stationarity, model minimality, descent of a direction, and decrease of a full step;
- state and prove the local quadratic-convergence theorem for Newton's method;
- classify the quadratic model when its Hessian is singular or indefinite;
- derive a shifted Newton direction and explain its steepest-descent limit; and
- distinguish exact line search from Armijo backtracking line search, use a shifted direction safeguard, and state when the unit Newton step is eventually accepted.

Chapter 8 described the common iterative framework and the one-dimensional subproblem created by line search. We now concentrate on how second-order information generates a multivariate search direction. Newton's method is locally extremely fast, but its Taylor model need not be convex, need not majorize the objective, and can be unreliable far from the target. These limitations motivate two different safeguards: modifying the model matrix and shortening the step by line search. The same model viewpoint will lead to steepest descent in Chapter 10.

9.1 The quadratic model and Newton equation

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and $f : \mathcal{D} \rightarrow \mathbb{R}$ be twice continuously differentiable. At $\mathbf{x}^{(k)} \in \mathcal{D}$, write

$$\mathbf{g}^{(k)} := \nabla f(\mathbf{x}^{(k)}), \quad \mathbf{H}^{(k)} := \mathbf{H}_f(\mathbf{x}^{(k)}).$$

For a displacement $\mathbf{d}$, the second-order Taylor model is

$$m^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{H}^{(k)} \mathbf{d}. \quad (9.1)$$

Its stationarity equation is

$$\mathbf{H}^{(k)} \mathbf{d} = -\mathbf{g}^{(k)}. \quad (9.2)$$

Thus Newton first asks for a stationary point of the quadratic truncation. It asks for the minimizer of that model only when the curvature makes this interpretation valid.

Definition 9.1 (Newton direction and full Newton step). If $\mathbf{H}^{(k)}$ is nonsingular, the unique solution $\mathbf{d}_N^{(k)}$ of

$$\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$$

is the *Newton direction*. The *full Newton step* sets

$$\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}.$$

Important note: Solve; do not invert

The identity $\mathbf{d}_N^{(k)} = -(\mathbf{H}^{(k)})^{-1} \mathbf{g}^{(k)}$ is useful in analysis. Numerically, solve the linear system in Equation (9.2); do not explicitly form the inverse.

Algorithm: Full-step Newton method

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, a stationarity tolerance $\varepsilon_g > 0$, and a maximum number K of updates. For $k = 0, \dots, K - 1$:

1. Compute $\mathbf{g}^{(k)} := \nabla f(\mathbf{x}^{(k)})$. If $\|\mathbf{g}^{(k)}\| \leq \varepsilon_g$, return $\mathbf{x}^{(k)}$.
2. Form $\mathbf{H}^{(k)} := \mathbf{H}_f(\mathbf{x}^{(k)})$ if necessary. If the Newton system is singular or numerically unreliable, stop with a diagnostic.
3. Solve $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$.
4. If $\mathbf{x}^{(k)} + \mathbf{d}_N^{(k)} \notin \mathcal{D}$, stop with a domain diagnostic; otherwise set $\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}$.

At $\mathbf{x}^{(K)}$, check the stopping test once more; otherwise report the update limit, not convergence.

For a dense $n \times n$ Hessian, forming it may require $O(n^2)$ storage and a dense factorization costs $O(n^3)$ arithmetic operations. Structure, sparsity, and Hessian-vector products can change these costs substantially. One should therefore compare derivative evaluations, linear solves, storage, and globalization trials rather than outer iterations alone.

9.2 Local quadratic convergence

For matrices, use the Euclidean induced operator norm

$$\|\mathbf{A}\|_2 := \max_{\|\mathbf{v}\|=1} \|\mathbf{A}\mathbf{v}\|.$$

It satisfies $\|\mathbf{A}\mathbf{v}\| \leq \|\mathbf{A}\|_2 \|\mathbf{v}\|$ and $\|\mathbf{A}\mathbf{B}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_2$.

Theorem 9.1 (Local quadratic convergence of Newton's method). *Let f be twice continuously differentiable near a stationary point $\mathbf{x}^*$. Assume that*

- (i) $\mathbf{H}^* := \mathbf{H}_f(\mathbf{x}^*)$ is nonsingular; and
- (ii) the Hessian is Lipschitz continuous near $\mathbf{x}^*$: there are $L > 0$ and $\delta_0 > 0$ such that

$$\|\mathbf{H}_f(\mathbf{x}) - \mathbf{H}_f(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|$$

whenever $\mathbf{x}, \mathbf{y} \in B(\mathbf{x}^, \delta_0)$.*

Then there are $\delta > 0$ and $C > 0$, with $B(\mathbf{x}^, \delta) \subset \mathcal{D}$, such that, whenever $\mathbf{x}^{(0)} \in B(\mathbf{x}^*, \delta)$,*

the full Newton iterates are well defined, remain in the ball, and satisfy

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^*\| \leq C \|\mathbf{x}^{(k)} - \mathbf{x}^*\|^2. \quad (9.3)$$

Thus the iterates converge to $\mathbf{x}^$, with at least Q -quadratic convergence (unless they reach the target in finitely many steps).*

Proof. Let

$$\mathbf{e}^{(k)} := \mathbf{x}^{(k)} - \mathbf{x}^*, \quad \mathbf{H}^{(k)} := \mathbf{H}_f(\mathbf{x}^{(k)}).$$

Because $\nabla f(\mathbf{x}^*) = \mathbf{0}$, the fundamental theorem of calculus along the segment from $\mathbf{x}^*$ to $\mathbf{x}^{(k)}$ gives

$$\nabla f(\mathbf{x}^{(k)}) = \left(\int_0^1 \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)}) dt \right) \mathbf{e}^{(k)}. \quad (9.4)$$

Combining the full Newton update with Equation (9.4) yields the key error identity

$$\mathbf{e}^{(k+1)} = (\mathbf{H}^{(k)})^{-1} \left(\int_0^1 [\mathbf{H}^{(k)} - \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)})] dt \right) \mathbf{e}^{(k)}. \quad (9.5)$$

Matrix inversion is continuous on the nonsingular matrices. After shrinking the neighborhood if necessary, every Hessian in it is nonsingular and

$$\|\mathbf{H}_f(\mathbf{x})^{-1}\|_2 \leq 2 \|(\mathbf{H}^*)^{-1}\|_2.$$

The Lipschitz condition gives, for $t \in [0, 1]$,

$$\|\mathbf{H}^{(k)} - \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)})\|_2 \leq L(1-t) \|\mathbf{e}^{(k)}\|.$$

Taking norms in Equation (9.5) and integrating,

$$\|\mathbf{e}^{(k+1)}\| \leq L \|(\mathbf{H}^*)^{-1}\|_2 \|\mathbf{e}^{(k)}\|^2.$$

Set $C := L \|(\mathbf{H}^*)^{-1}\|_2$. Choose $\delta > 0$ within the preceding neighborhood and so small that $C\delta \leq 1/2$. The quadratic bound then gives $\|\mathbf{e}^{(k+1)}\| \leq \frac{1}{2} \|\mathbf{e}^{(k)}\| < \delta$. Induction proves that all iterates are well defined, remain in the ball, and converge to $\mathbf{x}^*$; the sharper bound Equation (9.3) proves the stated rate. $\square$

Warning

This is a local root-finding theorem. Nonsingularity only at the target is enough because continuity then makes nearby Hessians nonsingular; it does not make a remote run well defined. Positive definiteness is not needed for quadratic convergence to a stationary point: the target may be a minimizer, a maximizer, or a saddle. The double-well example below demonstrates the last possibility. A small gradient is a stationarity test, not a certificate of minimality.

9.3 Computational examples: what pure Newton can do

9.3.1 A common coupled-quartic objective

We compare all the methods introduced in this chapter on

$$f(x_1, x_2) := (x_1^2 + x_2 - 1)^2 + (x_1 + x_2^2 - 1)^2, \quad \mathcal{D} = \mathbb{R}^2. \quad (9.6)$$

Write $a := x_1^2 + x_2 - 1$ and $b := x_1 + x_2^2 - 1$. Then

$$\nabla f(\mathbf{x}) = \begin{pmatrix} 4x_1a + 2b \\ 2a + 4x_2b \end{pmatrix}, \quad (9.7)$$

$$\mathbf{H}_f(\mathbf{x}) = \begin{pmatrix} 12x_1^2 + 4x_2 - 2 & 4(x_1 + x_2) \\ 4(x_1 + x_2) & 4x_1 + 12x_2^2 - 2 \end{pmatrix}. \quad (9.8)$$

The minimum value is zero. Indeed, $f \geq 0$, and its zeros satisfy $a = b = 0$. Subtracting these equations gives $(x_1 - x_2)(x_1 + x_2 - 1) = 0$. If $x_1 = x_2 = t$, then $t^2 + t - 1 = 0$; if $x_1 + x_2 = 1$, then $x_1^2 - x_1 = 0$. Consequently the complete set of global minimizers is

$$\left\{ (1, 0)^\top, (0, 1)^\top, (r_+, r_+)^\top, (r_-, r_-)^\top \right\}, \quad r_\pm := \frac{-1 \pm \sqrt{5}}{2}.$$

This multiplicity matters when comparing different initializations: all runs can succeed without reaching the same point.

From $\mathbf{x}^{(0)} = (1.23, 0.51)^\top$, pure Newton converges to $(1, 0)^\top$ in 16 updates with stopping test $\|\mathbf{g}^{(k)}\| \leq 10^{-8}$, but its objective values are not monotone. An illustration of the trajectory given by pure Newton is in [Figure 9.1](#). Selected iterate information are

k	$x_1^{(k)}$	$x_2^{(k)}$	$f(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $
0	1.230000	0.510000	1.286522	6.7402
2	0.921934	0.194982	0.003624	0.1038
3	0.811238	0.398298	0.004089	0.1388
4	0.399306	1.010676	0.205991	2.3251
5	0.287653	0.891603	0.007482	0.2786
6	3.647151	-0.959949	141.370900	172.8313
16	1.000000	$-1.03 \cdot 10^{-10}$	$2.20 \cdot 10^{-21}$	$4.98 \cdot 10^{-11}$

At $k = 5$, the Hessian eigenvalues are approximately $-8.35 \cdot 10^{-4}$ and 11.2503. Solving with the small negative eigenvalue amplifies the corresponding gradient component: $\|\mathbf{d}_N^{(5)}\| \approx 3.836$ and $(\mathbf{g}^{(5)})^\top \mathbf{d}_N^{(5)} \approx 0.005384 > 0$. The model value at the full trial is only 0.010174, whereas the actual value is 141.371. Thus this particular direction initially increases the function, and the long full step also travels far beyond the model's accurate region.

Warning

A negative Hessian eigenvalue does not by itself prove that the Newton direction

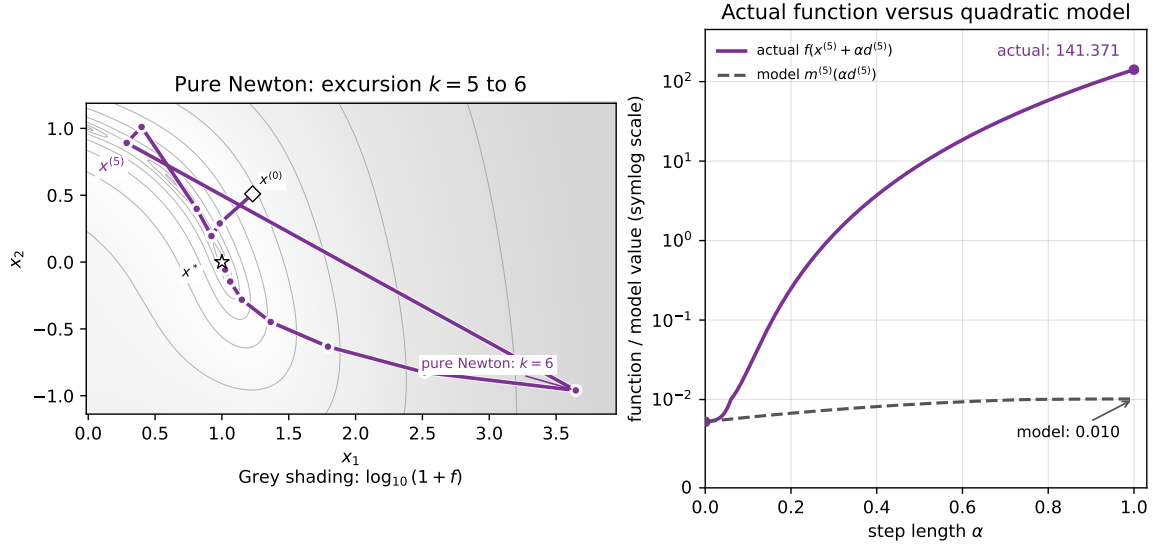


Figure 9.1: Pure Newton on Equation (9.6). The trajectory and local-model comparison illustrate the large excursion at update $k = 5$. The numerical explanation is given in the text, not inside the graph.

increases f . If $\mathbf{H} = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top$ is nonsingular and $\hat{\mathbf{g}} = \mathbf{U}^\top \mathbf{g}$, then

$$\mathbf{g}^\top \mathbf{d}_N = - \sum_{i=1}^n \frac{\hat{g}_i^2}{\lambda_i}.$$

The sign depends on the gradient components as well as the eigenvalues. For instance, $\mathbf{H} = \text{diag}(-1, 2)$ and $\mathbf{g} = (0, 1)^\top$ still give $\mathbf{g}^\top \mathbf{d}_N = -1/2 < 0$. A nearly zero eigenvalue only produces a large component if the gradient has a sufficient component in its eigendirection.

9.3.2 Convergence to a saddle on a double-well objective

Consider the double-well objective

$$F(x_1, x_2) := \frac{(x_1 + x_2)^4}{16} - x_1 x_2, \quad \mathcal{D} = \mathbb{R}^2, \quad \mathbf{x}^{(0)} = (0.4, -0.3)^\top. \quad (9.9)$$

The invertible change of coordinates $u = (x_1 + x_2)/2$, $v = (x_1 - x_2)/2$ gives

$$F = u^4 - u^2 + v^2 = (u^2 - \frac{1}{2})^2 + v^2 - \frac{1}{4}.$$

Hence the global minimizers are $\pm(1, 1)^\top/\sqrt{2}$, both with value $-1/4$. The only other stationary point is the origin; it is a saddle because $F(u, u) = u^4 - u^2 < 0$ for $0 < |u| < 1$, whereas $F(v, -v) = v^2 > 0$.

Writing $s = x_1 + x_2$, direct differentiation yields

$$\nabla F(\mathbf{x}) = \begin{pmatrix} s^3/4 - x_2 \\ s^3/4 - x_1 \end{pmatrix}, \quad \mathbf{H}_F(\mathbf{x}) = \begin{pmatrix} 3s^2/4 & 3s^2/4 - 1 \\ 3s^2/4 - 1 & 3s^2/4 \end{pmatrix}.$$

Its eigenvalues are $3s^2/2 - 1$ and 1. In particular, the Hessian at the saddle is nonsingular and indefinite.

The Newton equation transforms under this invertible linear change of coordinates into the separate Newton equations for $u^4 - u^2$ and v^2 . Indeed, if $\mathbf{x} = A(u, v)^\top$, then the transformed gradient and Hessian are $A^\top \nabla F$ and $A^\top \mathbf{H}_F A$; multiplying the transformed Newton equation by $(A^\top)^{-1}$ recovers the original one. Therefore

$$u^{(k+1)} = \frac{4(u^{(k)})^3}{6(u^{(k)})^2 - 1}, \quad v^{(k+1)} = 0.$$

Here $u^{(0)} = 0.05$, $v^{(0)} = 0.35$. Whenever $0 < |u| \leq 0.05$,

$$\left| \frac{u^+}{u} \right| = \frac{4u^2}{1 - 6u^2} \leq \frac{0.01}{0.985} < 1.$$

This interval is invariant, the denominator stays nonzero, and induction proves convergence to $u = 0$, not to either minimizer. Numerically,

$$\mathbf{x}^{(1)} = -\frac{1}{1970}(1, 1)^\top, \quad \mathbf{x}^{(2)} \approx 5.23 \cdot 10^{-10}(1, 1)^\top.$$

Thus the gradient tolerance 10^{-8} is reached after only two updates. See an illustration in Figure 9.2. Fast stationarity convergence can be the wrong outcome for minimization. We return to this same objective when studying shifted Newton.

9.4 Directions and decrease

Let $n \in \mathbb{N}$, let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, and let $f : \mathcal{D} \rightarrow \mathbb{R}$ be differentiable. A line-search iteration has the form

$$\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \alpha^{(k)} \geq 0. \quad (9.10)$$

The direction and the length have different roles: the direction should usually predict local decrease, while the step length decides how far that prediction can be trusted.

Definition 9.2 (Descent direction). Let $\mathbf{x} \in \mathcal{D}$. A vector $\mathbf{d} \in \mathbb{R}^n$ is a *strict descent direction* for f at $\mathbf{x}$ if

$$\nabla f(\mathbf{x})^\top \mathbf{d} < 0.$$

Proposition 9.2 (A descent direction decreases for short steps). *If f is differentiable at $\mathbf{x} \in \mathcal{D}$ and $\mathbf{d}$ is a strict descent direction there, then there is a $\delta > 0$ such that*

$$\mathbf{x} + \alpha \mathbf{d} \in \mathcal{D} \quad \text{and} \quad f(\mathbf{x} + \alpha \mathbf{d}) < f(\mathbf{x})$$

for every $\alpha \in (0, \delta]$.

Proof. Because $\mathcal{D}$ is open, $\mathbf{x} + \alpha \mathbf{d} \in \mathcal{D}$ for every sufficiently small $\alpha > 0$. Differentiability gives

$$f(\mathbf{x} + \alpha \mathbf{d}) = f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^\top \mathbf{d} + o(\alpha) \quad (\alpha \downarrow 0).$$

Let $c := -\nabla f(\mathbf{x})^\top \mathbf{d} > 0$. For all sufficiently small $\alpha > 0$, the remainder satisfies $|o(\alpha)| \leq c\alpha/2$. Hence

$$f(\mathbf{x} + \alpha \mathbf{d}) - f(\mathbf{x}) \leq -c\alpha + \frac{1}{2}c\alpha < 0.$$

Shrinking the upper bound on α to remain in $\mathcal{D}$ proves the claim. $\square$

Most line-search methods use descent directions, but the concept is not a universal requirement for every optimization method. Trust-region methods, negative-curvature steps, and algorithms designed to escape saddle points may select a step for reasons not summarized by the first directional derivative alone.

Proposition 9.3 (Positive curvature and the Newton direction). *If $\mathbf{H}^{(k)} \succ 0$, then $\mathbf{d}_N^{(k)}$ uniquely minimizes $m^{(k)}$. If in addition $\mathbf{g}^{(k)} \neq \mathbf{0}$, then it is a strict descent direction for f at $\mathbf{x}^{(k)}$.*

Proof. Positive definiteness makes $m^{(k)}$ strictly convex, so its unique stationary point is its unique global minimizer. Moreover,

$$(\mathbf{g}^{(k)})^\top \mathbf{d}_N^{(k)} = -(\mathbf{g}^{(k)})^\top (\mathbf{H}^{(k)})^{-1} \mathbf{g}^{(k)} < 0,$$

because $(\mathbf{H}^{(k)})^{-1} \succ 0$ and $\mathbf{g}^{(k)} \neq \mathbf{0}$. □

Warning

Four statements must be kept separate.

- Nonsingularity of the current Hessian makes the Newton equation uniquely solvable.
- Positive definiteness is a convenient sufficient condition making the Newton direction descending for every nonzero current gradient. It is not necessary for a particular gradient.
- A descent direction guarantees decrease for sufficiently small positive steps, not necessarily for the full step $\alpha^{(k)} = 1$.
- To run full Newton from a given initialization, every visited Hessian must be nonsingular and every new point must remain in $\mathcal{D}$.

For example, the coupled-quartic run has a positive-definite Hessian at $k = 3$, but its next full step increases f from 0.004089 to 0.205991. We examine this trial alongside shifted Newton below.

9.5 What a singular or indefinite Hessian means

Chapter 7 completely classified a convex quadratic. The same result explains what can go wrong with Newton's local model. Suppress the iteration superscript and consider

$$q(\mathbf{d}) := \mathbf{g}^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{H} \mathbf{d}, \quad \mathbf{H} = \mathbf{H}^\top. \quad (9.11)$$

Theorem 9.4 (Classification of the Newton quadratic model). *The model in Equation (9.11) has the following alternatives.*

(i) *If $\mathbf{H} \succeq 0$ and $\mathbf{g} \in \text{Range}(\mathbf{H})$, then*

$$\arg \min_{\mathbf{d} \in \mathbb{R}^n} q(\mathbf{d}) = \mathbf{d}_0 + \text{Null}(\mathbf{H}), \quad \mathbf{H} \mathbf{d}_0 = -\mathbf{g},$$

where $\mathbf{d}_0$ is any solution of the displayed linear system. If $\mathbf{H}$ is singular, this is an

unbounded affine set.

(ii) If $\mathbf{H} \succeq 0$ and $\mathbf{g} \notin \text{Range}(\mathbf{H})$, then q is unbounded below.

(iii) If $\mathbf{H}$ has a negative eigenvalue, then q is unbounded below.

Proof. For (i), since $\mathbf{g} \in \text{Range}(\mathbf{H})$, choose a solution $\mathbf{d}_0$ of $\mathbf{H}\mathbf{d}_0 = -\mathbf{g}$. For every $\mathbf{d}$,

$$q(\mathbf{d}) - q(\mathbf{d}_0) = \frac{1}{2}(\mathbf{d} - \mathbf{d}_0)^\top \mathbf{H}(\mathbf{d} - \mathbf{d}_0) \geq 0.$$

To identify equality, diagonalize the positive-semidefinite matrix: its quadratic form is a sum of nonnegative eigenvalues times squared coordinates, and vanishes exactly on its null space. Thus equality holds exactly when $\mathbf{d} - \mathbf{d}_0 \in \text{Null}(\mathbf{H})$, proving the formula. If $\mathbf{H}$ is singular, its null space contains a nonzero vector, and the affine solution set contains a line.

For (ii), the spectral theorem identifies the range of a symmetric matrix with the span of its nonzero-eigenvalue eigenvectors, and its null space with the span of the remaining ones. Hence $\text{Range}(\mathbf{H}) = \text{Null}(\mathbf{H})^\perp$. Hence there is a $\mathbf{z} \in \text{Null}(\mathbf{H})$ with $\mathbf{g}^\top \mathbf{z} \neq 0$. Along $\mathbf{d} = t\mathbf{z}$,

$$q(t\mathbf{z}) = t\mathbf{g}^\top \mathbf{z}.$$

Choosing the sign of t and letting $|t| \rightarrow \infty$ sends this value to $-\infty$.

For (iii), let $\mathbf{v} \neq \mathbf{0}$ satisfy $\mathbf{v}^\top \mathbf{H}\mathbf{v} < 0$. Then

$$q(t\mathbf{v}) = t\mathbf{g}^\top \mathbf{v} + \frac{1}{2}t^2\mathbf{v}^\top \mathbf{H}\mathbf{v} \longrightarrow -\infty$$

as $t \rightarrow +\infty$, because the negative quadratic term dominates the linear term. $\square$

For example, with $\mathbf{H} = \text{diag}(0, 1)$, the choice $\mathbf{g} = (0, 1)^\top$ gives the unbounded line of model minimizers $\{(t, -1)^\top : t \in \mathbb{R}\}$. The choice $\mathbf{g} = (1, 0)^\top$ gives no model minimizer at all. Thus a singular Hessian does not merely cause a numerical inconvenience: it records a flat model direction, and the linear term decides whether that direction produces nonuniqueness or unbounded descent.

9.6 Shifted Newton

One way to repair an unsuitable Hessian is to replace it by

$$\mathbf{B}^{(k)} := \mathbf{H}^{(k)} + \tau^{(k)}\mathbf{I}, \quad \tau^{(k)} \geq 0, \quad (9.12)$$

and solve

$$\mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}. \quad (9.13)$$

For a general objective this is most precisely called a *shifted* or *regularized* Newton step.

Proposition 9.5 (Descent and the steepest-descent limit). *Let $\mathbf{H}$ be symmetric and $\mathbf{g} \neq \mathbf{0}$. For every $\tau > -\lambda_{\min}(\mathbf{H})$, the matrix $\mathbf{H} + \tau\mathbf{I}$ is positive definite, and*

$$\mathbf{d}_\tau := -(\mathbf{H} + \tau\mathbf{I})^{-1}\mathbf{g}$$

is a strict descent direction. Moreover,

$$\tau \mathbf{d}_\tau \longrightarrow -\mathbf{g} \quad \text{as } \tau \rightarrow \infty. \quad (9.14)$$

Proof. By the spectral theorem, the eigenvalues of $\mathbf{H} + \tau I$ are $\lambda_i(\mathbf{H}) + \tau$, so the stated inequality makes all of them positive. Its inverse is therefore positive definite, and

$$\mathbf{g}^\top \mathbf{d}_\tau = -\mathbf{g}^\top (\mathbf{H} + \tau I)^{-1} \mathbf{g} < 0.$$

Let $\mathbf{H} = \mathbf{U} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top$ be an orthogonal eigendecomposition. Then

$$\tau(\mathbf{H} + \tau I)^{-1} = \mathbf{U} \operatorname{diag}\left(\frac{\tau}{\lambda_1 + \tau}, \dots, \frac{\tau}{\lambda_n + \tau}\right) \mathbf{U}^\top \rightarrow I.$$

Multiplying by $-\mathbf{g}$ proves Equation (9.14). $\square$

The limit concerns direction: the normalized $\mathbf{d}_\tau$ approaches the normalized negative gradient. The unscaled vector itself tends to zero. Thus a large shift makes the method more conservative and increasingly steepest-descent-like.

If $\mathbf{H}$ is nonsingular, continuity of inversion also gives $\mathbf{d}_\tau \rightarrow -\mathbf{H}^{-1} \mathbf{g}$ as $\tau \downarrow 0$, wherever the inverse is defined. This algebraic limit need not stay within the positive-definite shifted family when $\mathbf{H}$ is indefinite.

9.6.1 Convex quadratic models and majorization

More generally, a symmetric model matrix $\mathbf{B}^{(k)}$ defines

$$M^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{B}^{(k)} \mathbf{d}. \quad (9.15)$$

If $\mathbf{B}^{(k)} \succ 0$, its unique minimizer is $\mathbf{d}_M^{(k)} = -(\mathbf{B}^{(k)})^{-1} \mathbf{g}^{(k)}$. We say that this model *majorizes* f locally at $\mathbf{x}^{(k)}$ if there is a neighborhood U of $\mathbf{0}$, with $\mathbf{x}^{(k)} + U \subseteq \mathcal{D}$, such that

$$f(\mathbf{x}^{(k)} + \mathbf{u}) \leq M^{(k)}(\mathbf{u}) \quad (\mathbf{u} \in U), \quad M^{(k)}(\mathbf{0}) = f(\mathbf{x}^{(k)}).$$

Proposition 9.6 (Local decrease from majorization). *Suppose $\mathbf{B}^{(k)} \succ 0$, $\mathbf{g}^{(k)} \neq \mathbf{0}$, and the local majorization inequality above holds. Then*

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}_M^{(k)}) < f(\mathbf{x}^{(k)}) \quad \text{whenever } 0 < \alpha \leq 1 \text{ and } \alpha \mathbf{d}_M^{(k)} \in U.$$

In particular, every sufficiently small positive step decreases f .

Proof. The strictly convex model has unique minimizer $\mathbf{d}_M^{(k)} \neq \mathbf{0}$, so $M^{(k)}(\mathbf{d}_M^{(k)}) < M^{(k)}(\mathbf{0})$. Majorization and convexity give

$$\begin{aligned} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}_M^{(k)}) &\leq M^{(k)}(\alpha \mathbf{d}_M^{(k)}) \\ &\leq (1 - \alpha) M^{(k)}(\mathbf{0}) + \alpha M^{(k)}(\mathbf{d}_M^{(k)}) \\ &< M^{(k)}(\mathbf{0}) = f(\mathbf{x}^{(k)}). \end{aligned}$$

Since U is a neighborhood of $\mathbf{0}$, the required inclusion holds for every sufficiently small $\alpha > 0$. $\square$

The full step is covered only if $\mathbf{d}_M^{(k)} \in U$. Positive definiteness alone does not establish an upper bound covering the full step, as the counterexamples in Section 9.6.3 illustrate. Chapter 10 develops the choice $\mathbf{B}^{(k)} = (1/t^{(k)})I$ and its majorization guarantee (Lemma 10.2 and proposition 10.3); other model matrices lead to variable-metric methods, which are outside this course.

9.6.2 Choosing the shift without a function-decrease test

Proposition 9.7 (A sufficiently large shift gives a decreasing full trial). *Let f be differentiable at a nonstationary point $\mathbf{x} \in \mathcal{D}$, and let $\mathbf{H}$ be any fixed symmetric matrix. Then, for all sufficiently large τ , the shifted trial*

$$\mathbf{x} + \mathbf{d}_\tau, \quad \mathbf{d}_\tau = -(\mathbf{H} + \tau \mathbf{I})^{-1} \nabla f(\mathbf{x}),$$

belongs to $\mathcal{D}$ and has smaller objective value than $\mathbf{x}$.

Proof. By Proposition 9.5, $\tau \mathbf{d}_\tau \rightarrow -\nabla f(\mathbf{x})$, so $\mathbf{d}_\tau \rightarrow \mathbf{0}$ and the trial eventually belongs to the open set $\mathcal{D}$. Differentiability gives

$$f(\mathbf{x} + \mathbf{d}_\tau) - f(\mathbf{x}) = \nabla f(\mathbf{x})^\top \mathbf{d}_\tau + o(\|\mathbf{d}_\tau\|).$$

Multiplying by τ , the first term converges to $-\|\nabla f(\mathbf{x})\|^2 < 0$. Also $\tau \|\mathbf{d}_\tau\| \rightarrow \|\nabla f(\mathbf{x})\|$, so $\tau o(\|\mathbf{d}_\tau\|) \rightarrow 0$. The scaled objective difference is therefore negative for every sufficiently large τ , proving the claim. $\square$

Algorithm: Shifted Newton with full steps

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\varepsilon_g > 0$, a positive seed $\tau_{\min}$, a growth factor $\gamma > 1$, and a maximum number K of updates. For $k = 0, \dots, K - 1$:

1. Compute $\mathbf{g}^{(k)}$; return if $\|\mathbf{g}^{(k)}\| \leq \varepsilon_g$. Form $\mathbf{H}^{(k)}$.
2. Test shifts $0, \tau_{\min}, \gamma\tau_{\min}, \gamma^2\tau_{\min}, \dots$, and take the first $\tau^{(k)}$ for which $\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau^{(k)} \mathbf{I} \succ \mathbf{0}$.
3. Solve $\mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}$.
4. Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}_M^{(k)}$. If this point is outside $\mathcal{D}$, report a domain failure.

At $\mathbf{x}^{(K)}$, check the stopping test once more; otherwise report the update limit. Numerical failures must be reported separately from convergence.

The shift search terminates in exact arithmetic: its trial values grow without bound, so eventually $\tau > -\lambda_{\min}(\mathbf{H}^{(k)})$. This only proves termination of the matrix-repair step, not convergence of the outer iteration. Our experiments use $\tau_{\min} = 0.1$ and $\gamma = 2$, resetting the first trial to zero at each new iterate.

Warning

The algorithm accepts the *first positive-definite model matrix*; it does not test whether the full step decreases f . Proposition 9.7 says that a sufficiently large shift would give decrease, not that the first admissible shift is large enough. Increasing the shift until the full trial also decreases f would define a different algorithm. The shifted Newton method here can be nonmonotone. Nor does positive definiteness alone ensure that a full trial stays in a general domain $\mathcal{D}$.

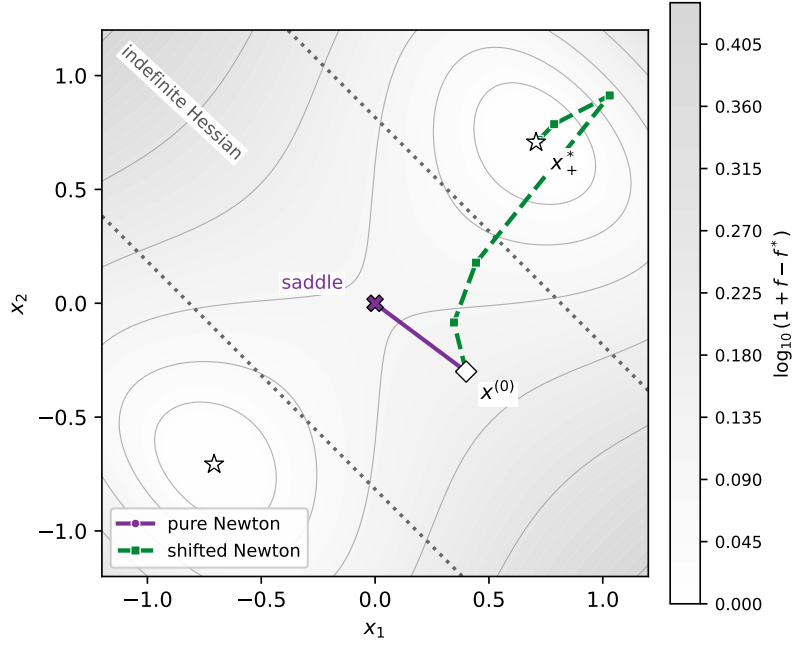


Figure 9.2: The same double-well objective and initialization for both methods. Shifting changes the limiting point from the saddle to a minimizer. This is an example, not a guarantee that shifting avoids every saddle.

9.6.3 What shifting helps, and what it does not

On the double-well problem [Equation \(9.9\)](#), the initial Hessian has eigenvalues -0.985 and 1 . The trial shifts $0, 0.1, 0.2, 0.4, 0.8$ fail; 1.6 is the first admissible value. The positive shifts during the run are $1.6, 1.6, 0.8$, after which no shift is needed. Shifted Newton reaches $(1, 1)^T / \sqrt{2}$ in eight updates; pure Newton reaches the saddle in two. See the trajectories in [Figure 9.2](#). Thus the common gradient stopping test must be interpreted together with the limiting point's curvature.

On the coupled-quartic objective [Equation \(9.6\)](#), shifted Newton takes eight updates, but its step from $k = 2$ to $k = 3$ increases f . The following two trials separate model convexity from full-step decrease. Here $\mathbf{B}^{(k)} = \mathbf{H}^{(k)}$ for pure Newton and $\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau^{(k)}I$ for shifted Newton.

	pure Newton, $k = 3$	shifted Newton, $k = 2$
$\tau^{(k)}$	0	0.1
$\lambda_{\min}(\mathbf{B}^{(k)})$	0.016684	0.036669
$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}$	-0.010886	-0.006788
$f(\mathbf{x}^{(k)})$	0.004089	0.003624
$M^{(k)}(\mathbf{d}^{(k)})$	-0.001354	0.000230
$f(\mathbf{x}^{(k)} + \mathbf{d}^{(k)})$	0.205991	0.014592

Both positive-definite models give descent directions, yet their full steps increase the objective. In the shifted run, the direction comes from $\mathbf{B}^{(2)}$, not the indefinite raw Hessian; step-length control addresses the remaining failure.

9.7 Newton with line search

Line search keeps a chosen direction fixed and changes its scalar length. It is a tool for obtaining convergence guarantees under suitable assumptions, not a necessary ingredient of every convergent Newton run. For a direction $\mathbf{d}^{(k)}$, define

$$\mathcal{I}^{(k)} := \{\alpha \geq 0 : \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)} \in \mathcal{D}\}.$$

When the minimum is attained, exact line search chooses

$$\alpha^{(k)} \in \arg \min_{\alpha \in \mathcal{I}^{(k)}} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}). \quad (9.16)$$

Warning

The argmin in Equation (9.16) may be empty. For $f(x) = e^{-x}$, $x^{(k)} = 0$, and $d^{(k)} = 1$, the direction descends and every positive step is feasible, but the infimum along the ray is not attained. Solving $(d/d\alpha)f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = 0$ also finds only stationary candidates for the line problem. Likewise, Chapter 8's one-dimensional methods are not automatic global solvers of an arbitrary line restriction: derivative bisection needs a sign bracket and additional minimality information, Newton is local, and golden-section search needs unimodality on a containing bracket.

An inexact search is usually cheaper. Define $\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$. The Armijo condition compares this line restriction with $\ell^{(k)}(\alpha) = f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}$, asking for a fixed fraction of the decrease predicted by the first-order model. Acceptable steps lie where $\phi^{(k)} \leq \ell^{(k)}$. There may be several disjoint acceptable intervals: backtracking need not locate their endpoints or the largest acceptable step.

Definition 9.3 (Armijo condition). Let $\mathbf{d}^{(k)}$ be a strict descent direction. For $\sigma \in (0, 1)$, a positive step α satisfies the Armijo condition if

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}. \quad (9.17)$$

Proposition 9.8 (Finite termination of Armijo backtracking). *Suppose f is differentiable at $\mathbf{x} \in \mathcal{D}$ and $\nabla f(\mathbf{x})^\top \mathbf{d} < 0$. For every $\bar{\alpha} > 0$, $\beta \in (0, 1)$, and $\sigma \in (0, 1)$, one of the trials $\bar{\alpha}, \beta \bar{\alpha}, \beta^2 \bar{\alpha}, \dots$ is feasible and satisfies the Armijo condition.*

Proof. Differentiability gives

$$f(\mathbf{x} + \alpha \mathbf{d}) = f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^\top \mathbf{d} + o(\alpha).$$

Put $c := -\nabla f(\mathbf{x})^\top \mathbf{d} > 0$. For every sufficiently small $\alpha > 0$, openness of $\mathcal{D}$ makes the trial feasible and $|o(\alpha)| \leq (1 - \sigma)c\alpha$. Hence

$$f(\mathbf{x} + \alpha \mathbf{d}) \leq f(\mathbf{x}) - c\alpha + (1 - \sigma)c\alpha = f(\mathbf{x}) + \sigma \alpha \nabla f(\mathbf{x})^\top \mathbf{d}.$$

Since $\beta^r \bar{\alpha} \rightarrow 0$, the backtracking sequence eventually enters this interval of acceptable steps. $\square$

Algorithm: Armijo backtracking line search

Choose $\bar{\alpha} > 0$, $\beta \in (0, 1)$, and $\sigma \in (0, 1)$. For $r = 0, 1, 2, \dots$, set $\alpha = \beta^r \bar{\alpha}$. Return the first feasible trial satisfying Equation (9.17). Starting at $r = 0$ ensures that $\bar{\alpha}$ itself is tested.

Algorithm: Newton–Armijo with a shifted direction safeguard

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\varepsilon_g > 0$, a maximum number K of updates, shift parameters $\tau_{\min} > 0, \gamma > 1$, and Armijo parameters $\bar{\alpha} = 1$, $\beta, \sigma \in (0, 1)$. For $k = 0, \dots, K - 1$:

1. Compute $\mathbf{g}^{(k)}$; return if $\|\mathbf{g}^{(k)}\| \leq \varepsilon_g$. Form $\mathbf{H}^{(k)}$.
2. Try solving $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$. If the solve succeeds with a finite direction and $(\mathbf{g}^{(k)})^\top \mathbf{d}_N^{(k)} < 0$, set $\mathbf{d}^{(k)} = \mathbf{d}_N^{(k)}$ and $\tau^{(k)} = 0$.
3. Otherwise, test $\tau = \tau_{\min}, \gamma\tau_{\min}, \dots$. For each trial, attempt to solve

$$(\mathbf{H}^{(k)} + \tau \mathbf{I}) \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}.$$

Accept the first successful finite solve satisfying $(\mathbf{g}^{(k)})^\top \mathbf{d}_M^{(k)} < 0$, and set $\mathbf{d}^{(k)} = \mathbf{d}_M^{(k)}$, $\tau^{(k)} = \tau$. No positive-definiteness test is imposed on this shifted matrix.

4. Keeping this direction fixed, use Armijo backtracking line search to obtain $\alpha^{(k)}$.
5. Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}$.

Check the stopping test again after the last permitted update. Report numerical failures or an update limit, rather than calling them convergence.

Important note: Test the direction, not every direction

Once the linear system has been solved, the scalar test $\mathbf{g}^\top \mathbf{d} < 0$ requires only an inner product, costing $O(n)$ arithmetic operations. It certifies descent for this particular gradient and direction. Positive definiteness is stronger: it guarantees descent for every nonzero gradient, and therefore need not be required for line search. For example, $\mathbf{B} = \text{diag}(-1, 2)$ is indefinite, but with $\mathbf{g} = (0, 1)^\top$ its solution is $\mathbf{d} = (0, -1/2)^\top$, giving $\mathbf{g}^\top \mathbf{d} = -1/2 < 0$.

The descent-only safeguard avoids a separate positive-definiteness check. Such a check does *not* require all eigenvalues: for a symmetric matrix, Cholesky factorization is another standard option and may also be reused for the solve when it succeeds. The practical cost comparison depends on the solver, structure, and number of rejected shifts; descent-only testing is not automatically faster in every implementation.

Proposition 9.9 (Finite termination of the shifted direction search). *At a nonstationary point, let $\mathbf{g} \neq \mathbf{0}$ and let $\mathbf{H}$ be symmetric. For $\tau_{\min} > 0$ and $\gamma > 1$, the shifted direction search above accepts a finite trial in exact arithmetic.*

Proof. For $\tau > \|\mathbf{H}\|_2$ and every $\mathbf{v} \neq \mathbf{0}$,

$$\mathbf{v}^\top (\mathbf{H} + \tau \mathbf{I}) \mathbf{v} \geq (\tau - \|\mathbf{H}\|_2) \|\mathbf{v}\|^2 > 0.$$

Thus $\mathbf{B} = \mathbf{H} + \tau \mathbf{I}$ is positive definite and the system $\mathbf{B} \mathbf{d} = -\mathbf{g}$ has a unique nonzero solution.

This solution satisfies $\mathbf{g}^\top \mathbf{d} = -\mathbf{d}^\top \mathbf{B} \mathbf{d} < 0$, so the trial is accepted. Since $\gamma^r \tau_{\min} \rightarrow \infty$, the tested shifts eventually exceed $\|\mathbf{H}\|_2$. Positive definiteness is used here only to prove that some trial will work; an earlier indefinite trial may already be accepted. $\square$

Warning

The two shifting rules in this chapter have different roles. Full-step shifted Newton requires a positive-definite model matrix. Newton–Armijo requires only a solvable shifted equation and a strict descent direction; Armijo backtracking then controls the step length. If the accepted matrix is indefinite, the direction is a stationary point of an unbounded-below quadratic model, not its minimizer. The line-search guarantee uses the directional derivative, not minimization or convexity of that model.

The fallback is mandatory when the raw direction is unavailable or fails the descent test; simply backtracking an ascending Newton direction does not satisfy the hypotheses of [Proposition 9.8](#). For example, $f(x) = -x^2/2$ at $x = 1$ has Newton direction $d = -1$, so $f'(1)d = 1 > 0$. For every $\sigma < 1$, all sufficiently small positive steps fail Armijo because $f(1 - \alpha) - f(1) = \alpha - \alpha^2/2 > \sigma\alpha$. In contrast, an indefinite Hessian whose Newton direction happens to descend need not be shifted by this algorithm.

One may replace the backtracking step with exact line search, provided the line minimum exists and can be found. The supplied Python implementation uses Armijo backtracking line search, not an exact global line solver.

Theorem 9.10 (The unit Newton step is eventually accepted). *Let f be twice continuously differentiable near $\mathbf{x}^* \in \mathcal{D}$, with locally Lipschitz Hessian, $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and $\mathbf{H}_f(\mathbf{x}^*) \succ 0$. Use the safeguarded Newton–Armijo method with $\bar{\alpha} = 1$ and $\sigma \in (0, 1/2)$. If its iterates converge to $\mathbf{x}^*$, then, at all sufficiently late nonterminal iterations, the raw Newton direction is selected and $\alpha^{(k)} = 1$ is accepted. Moreover, every sufficiently close initialization generates a locally Q -quadratically convergent sequence, unless it reaches $\mathbf{x}^*$ in finitely many steps.*

Proof. Continuity and positive definiteness give a ball about $\mathbf{x}^*$ and $m > 0$ such that $\mathbf{H}_f(\mathbf{x}) \succeq m\mathbf{I}$ throughout that ball. There the Newton direction exists and descends whenever $\nabla f(\mathbf{x}) \neq \mathbf{0}$, so the direction safeguard keeps it. Its norm satisfies $\|\mathbf{d}_N\| \leq \|\nabla f(\mathbf{x})\|/m \rightarrow 0$ as $\mathbf{x} \rightarrow \mathbf{x}^*$. After shrinking the neighborhood, the whole segment from $\mathbf{x}$ to $\mathbf{x} + \mathbf{d}_N$ lies in the ball.

Let L be a Lipschitz constant for the Hessian there. Applying the fundamental theorem of calculus twice gives

$$\begin{aligned} f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} &= \int_0^1 (1-t) \mathbf{d}^\top \mathbf{H}_f(\mathbf{x} + t\mathbf{d}) \mathbf{d} dt, \\ R(\mathbf{x}, \mathbf{d}) &:= f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} - \frac{1}{2} \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}) \mathbf{d}, \\ |R(\mathbf{x}, \mathbf{d})| &\leq L \|\mathbf{d}\|^3 \int_0^1 t(1-t) dt = \frac{L}{6} \|\mathbf{d}\|^3. \end{aligned}$$

For a Newton direction, $\mathbf{g}^\top \mathbf{d}_N = -\mathbf{d}_N^\top \mathbf{H}_f(\mathbf{x}) \mathbf{d}_N$. Consequently

$$\begin{aligned} f(\mathbf{x} + \mathbf{d}_N) - f(\mathbf{x}) - \sigma \mathbf{g}^\top \mathbf{d}_N &= -\left(\frac{1}{2} - \sigma\right) \mathbf{d}_N^\top \mathbf{H}_f(\mathbf{x}) \mathbf{d}_N + R(\mathbf{x}, \mathbf{d}_N) \\ &\leq -\left(\frac{1}{2} - \sigma\right) m \|\mathbf{d}_N\|^2 + \frac{L}{6} \|\mathbf{d}_N\|^3 \leq 0 \end{aligned}$$

whenever $\|\mathbf{d}_N\|$ is sufficiently small. Thus every point in a sufficiently small neighborhood accepts the first, unit trial. A convergent sequence eventually enters this neighborhood, proving the first assertion.

For the local-convergence assertion, shrink the invariant ball in [Theorem 9.1](#) to lie in this unit-acceptance neighborhood. Starting there, induction shows that every safeguarded iteration agrees with pure Newton, remains in the ball, and satisfies the quadratic error bound. This proves the claim without assuming convergence in advance. $\square$

Warning

The restriction $\sigma < 1/2$ is for eventual unit Newton steps, not for finite backtracking, which only needs $0 < \sigma < 1$. For $f(x) = x^2/2$ and a nonzero current point, the full Newton step reaches zero; its Armijo right-hand side is $(1/2 - \sigma)x^2$. It therefore fails whenever $\sigma > 1/2$.

Finite termination of each line search is not a proof of convergence of the outer algorithm from every start. Even strict descent and Armijo decrease alone do not force gradients to vanish. For $f(x) = x^2$, $x^{(0)} = 1$, choose $\mathbf{d}^{(k)} = -2^{-k-3}x^{(k)}$ and take unit steps. They satisfy Armijo with $\sigma = 1/4$, since $(1 - \epsilon)^2 \leq 1 - \epsilon/2$ for $\epsilon = 2^{-k-3} \leq 1/8$. Yet

$$x^{(k)} = \prod_{j=0}^{k-1} (1 - 2^{-j-3}) \geq 1 - \sum_{j=0}^{k-1} 2^{-j-3} \geq \frac{3}{4}.$$

(The product inequality follows by induction from $(1 - a)(1 - b) \geq 1 - a - b$ for $a, b \geq 0$.) The decreasing sequence has a nonstationary limit. This is not Newton's method, but shows why stronger global-convergence assumptions are needed.

9.8 Numerical comparisons and method selection

9.8.1 One objective, one start, three distinct behaviors

Return to [Equation \(9.6\)](#), with $\mathbf{x}^{(0)} = (1.23, 0.51)^\top$ and tolerance 10^{-8} . Both full-step methods increase the objective at $k = 2$. At that iterate the safeguarded Newton–Armijo method uses the same shifted direction as shifted Newton: $\tau^{(2)} = 0.1$ and $(\mathbf{g}^{(2)})^\top \mathbf{d}^{(2)} \approx -0.00678804$. With $\sigma = 10^{-4}$ and $\beta = 1/2$, its trials are

α	$f(\mathbf{x}^{(2)} + \alpha \mathbf{d}^{(2)})$	Armijo bound	decision
1	0.014591780	0.003623132	reject
1/2	0.000749134	0.003623471	accept

Only the length changes between these two trials. Unit steps are accepted at every later iteration $k \geq 3$.

All methods stop at the first recorded iterate with $\|\mathbf{g}^{(k)}\| \leq 10^{-8}$. Full-step shifted Newton tests shifts $0, 0.1, 0.2, 0.4, \dots$. Newton–Armijo first tests the raw direction, then tries $0.1, 0.2, 0.4, \dots$ only if needed; Newton–Armijo uses $\bar{\alpha} = 1$, $\beta = 1/2$, $\sigma = 10^{-4}$. Trajectories and convergence curves are collected in [Figures 9.3](#) and [9.4](#).

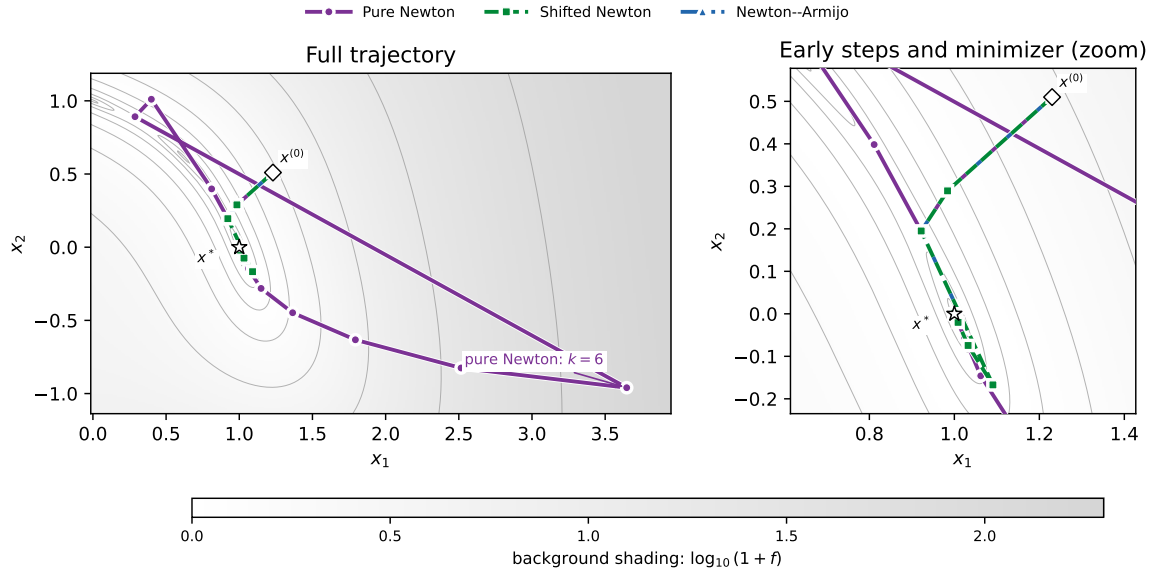


Figure 9.3: Three algorithms on the same coupled-quartic objective from $(1.23, 0.51)^\top$. All converge to $(1, 0)^\top$, but the raw Newton path makes a much larger excursion. Grayscale contours separate the background from the colored iteration paths.

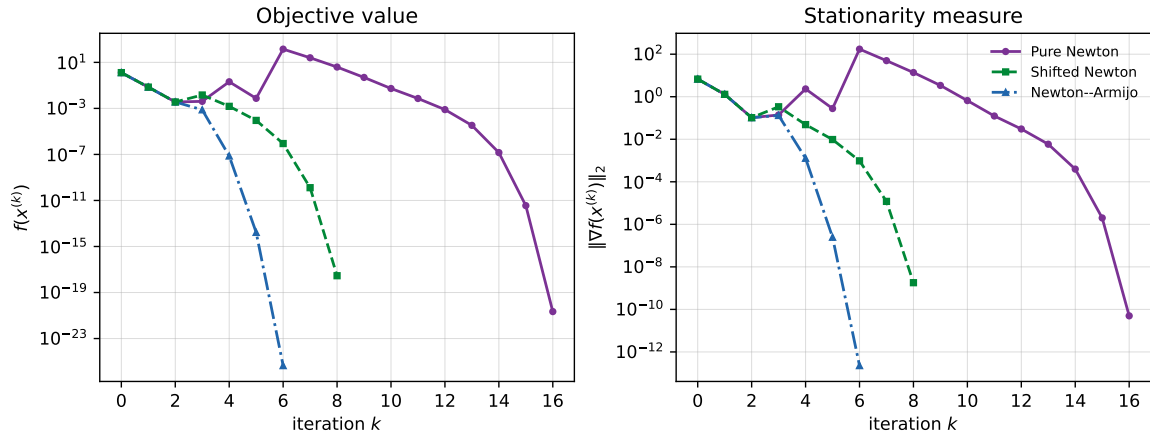


Figure 9.4: Objective values and gradient norms for the same three runs. Nonmonotonicity is compatible with eventual convergence. A small final objective and a small final gradient measure different aspects of a run.

Method	Updates	f calls	∇f calls	$\mathbf{H}_f$ calls	Solves
Pure Newton	16	17	17	16	16
Shifted Newton	8	9	9	8	8
Newton–Armijo	6	14	7	6	7

Their final gradient norms are approximately $4.98 \cdot 10^{-11}$, $1.80 \cdot 10^{-9}$, and $2.27 \cdot 10^{-13}$, respectively. Pure Newton has three objective increases (steps starting at $k = 2, 3, 5$); shifted Newton has one (at $k = 2$); Newton–Armijo has none.

The counters include the value and gradient at each recorded iterate, including the initial and terminal points. A *solve* is a completed linear-system solve, including a raw or shifted direction

later rejected as non-descending. This accounts for seven solves in six Newton–Armijo updates. Cholesky tests used by full-step shifted Newton and the diagnostic eigenvalue calculations are excluded from the solve count. Backtracking adds function evaluations at trial points, not gradient or Hessian evaluations at every trial. The history collection also evaluates f for pure and shifted Newton even though those algorithms do not need objective values to accept their steps.

These counts are reproducible implementation measurements, not problem-independent complexity bounds or wall-clock timings. Dense matrix factorizations, failed Cholesky tests, derivative costs, sparsity, storage, and hardware all matter in a larger problem.

9.8.2 Changing the initialization can reverse the ranking

Keep the objective, parameters, and gradient tolerance unchanged, but start at $\mathbf{x}^{(0)} = (0, 0.15)^\top$. The outcomes become

Method	Updates	f calls	Solves	Limiting global minimizer
Pure Newton	6	7	6	$(r_-, r_-)^\top$
Shifted Newton	9	10	9	$(0, 1)^\top$
Newton–Armijo	9	21	15	$(r_+, r_+)^\top$

All three achieve the global minimum value zero, although they converge to different global minimizers; see [Figure 9.5](#). Pure Newton is now the fastest in *number of updates*; its first objective value rises from 1.678006 to 5.832568, but this nonmonotone step does not prevent convergence. Shifted Newton also has a first-step increase, to about 3.722995. Newton–Armijo decreases at every accepted update.

Here the weaker safeguard makes a visible difference at $k = 0$. The raw Newton direction has $(\mathbf{g}^{(0)})^\top \mathbf{d}_N^{(0)} \approx 9.357685 > 0$. The shifted solves for $\tau = 0.1, 0.2, 0.4, 0.8$ also give ascending directions. At $\tau = 1.6$, however,

$$\mathbf{B}^{(0)} = \begin{pmatrix} 0.2 & 0.6 \\ 0.6 & -0.13 \end{pmatrix}, \quad \mathbf{d}^{(0)} \approx \begin{pmatrix} 4.212565 \\ 1.854145 \end{pmatrix}, \quad (\mathbf{g}^{(0)})^\top \mathbf{d}^{(0)} \approx -12.475067 < 0.$$

Its determinant is $-0.386 < 0$, so the accepted matrix is indefinite. Armijo rejects $\alpha = 1, 1/2$ and accepts $\alpha = 1/4$, giving $\mathbf{x}^{(1)} \approx (1.053141, 0.613536)^\top$ and $f(\mathbf{x}^{(1)}) \approx 0.706741$. In contrast, the full-step shifted Newton rule waits for $\tau = 3.2$ to obtain a positive-definite matrix. The Newton–Armijo run uses one further positive shift, $\tau^{(3)} = 0.1$, and accepts unit steps at every $k \geq 1$. Its fifteen completed solves include nine raw solves, five shifted solves at $k = 0$, and one at $k = 3$; the four rejected shifted directions still incur solve cost.

Warning

These runs do not establish a general ranking: update counts and plotted path lengths are not complete measures of computational cost.

9.8.3 A case where line search changes divergence into convergence

Consider the strictly convex function

$$h(x) := \sqrt{1 + x^2}, \quad \mathcal{D} = \mathbb{R}, \quad h'(x) = \frac{x}{\sqrt{1 + x^2}}, \quad h''(x) = \frac{1}{(1 + x^2)^{3/2}} > 0.$$

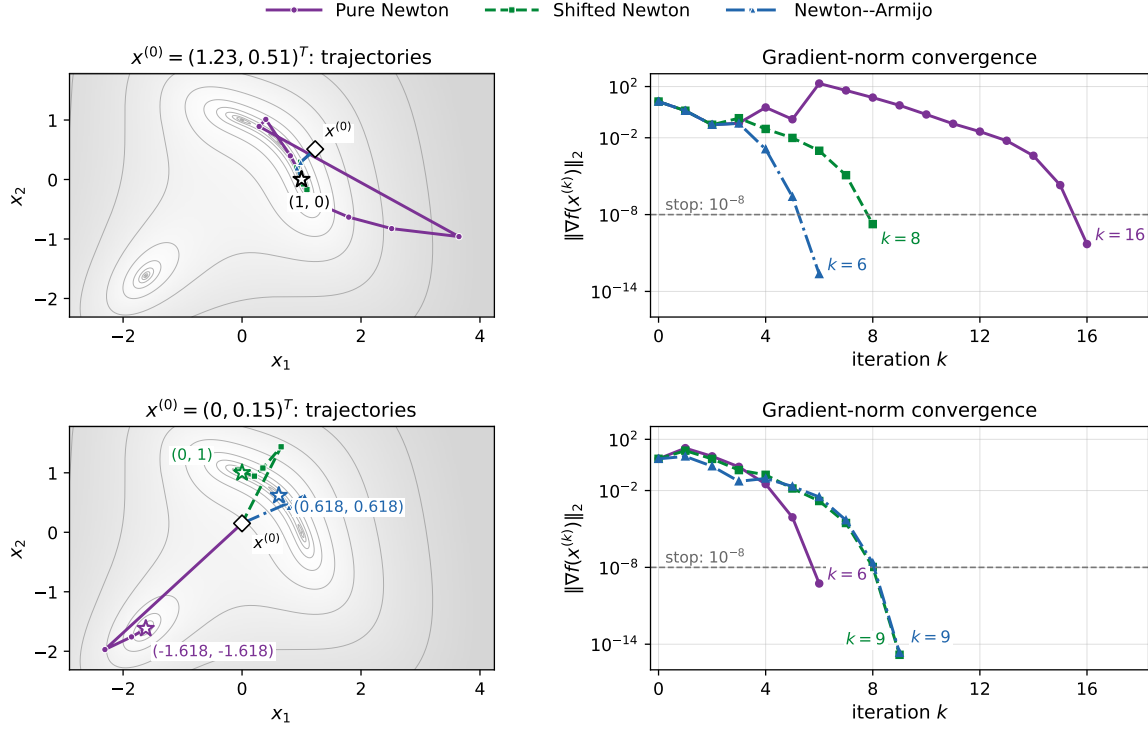


Figure 9.5: Two initializations, each shown by an iteration trajectory and a gradient-norm history. The starting point changes both the relative iteration counts and the minimizer reached. The common tolerance is 10^{-8} ; the two rows use shared plotting scales.

Its unique global minimizer is zero. Full Newton has the explicit recurrence

$$d^{(k)} = -x^{(k)}(1 + (x^{(k)})^2), \quad x^{(k+1)} = -(x^{(k)})^3.$$

Thus $|x^{(0)}| < 1$ gives convergence to zero, $|x^{(0)}| = 1$ gives a two-cycle, and $|x^{(0)}| > 1$ gives divergence because $|x^{(k)}| = |x^{(0)}|^{3^k}$. From $x^{(0)} = 2$, the first iterates are $2, -8, 512, -134217728, \dots$. All visited Hessians are positive; this failure does not require negative curvature. The PD-only shifting rule also uses zero shift here and therefore has the same failure.

For Newton-Armijo with $\bar{\alpha} = 1, \beta = 1/2, \sigma = 10^{-4}$, the initial direction is -10 . Trials 1 and $1/2$ are rejected; $1/4$ is accepted, giving $x^{(1)} = -1/2$. For example, the rejected $1/2$ trial has value $\sqrt{10} > \sqrt{5}$, whereas the accepted $1/4$ trial has value $\sqrt{5/4}$ and Armijo bound $\sqrt{5} - \sigma\sqrt{5}$.

To justify all subsequent unit steps, put $t = x^2 \leq 1/4$. A full Newton step gives $x^+ = -x^3$, and

$$h(x) - h(x^+) = \frac{t - t^3}{\sqrt{1+t} + \sqrt{1+t^3}}, \quad -h'(x)d = t\sqrt{1+t}.$$

For $t > 0$, the ratio of actual to predicted decrease is

$$\frac{1 - t^2}{\sqrt{1+t}(\sqrt{1+t} + \sqrt{1+t^3})} \geq \frac{15/16}{5/2} = \frac{3}{8} > 10^{-4}.$$

Here $t^3 \leq t$, so the denominator is at most $2(1+t) \leq 5/2$. Hence every unit step satisfies Armijo and stays in $|x| \leq 1/2$. The resulting sequence is

$$2, -\frac{1}{2}, \frac{1}{8}, -\frac{1}{512}, \frac{1}{134217728}, \dots \rightarrow 0.$$

The gradient tolerance 10^{-8} is reached after four updates; the last displayed point is approximately $7.45 \cdot 10^{-9}$, not exactly zero. This explicit example shows why shortening a Newton step can matter far from a minimizer, while the previous comparisons show why line search is not always necessary for a successful run.

Chapter summary

- Newton solves the stationarity equation of the second-order Taylor model; only positive curvature makes that point the unique model minimizer.
- A positive-definite Hessian makes the Newton direction descending, but does not guarantee decrease of the full step.
- A nonsingular limiting Hessian, a locally Lipschitz Hessian, and a close initialization yield local Q-quadratic convergence to a stationary point.
- A positive-semidefinite singular model Hessian produces either an affine family of minimizers or no minimizer, depending on the range condition; negative curvature makes the model unbounded below.
- Shifting repairs model curvature; taking the first positive-definite shift does not ensure full-step decrease. At a fixed point, $\tau \mathbf{d}_\tau \rightarrow -\mathbf{g}$.
- Armijo backtracking terminates for every descent direction; raw Newton needs a shifted fallback when its direction fails that test. This fallback checks a solve and $\mathbf{g}^\top \mathbf{d} < 0$, not positive definiteness. Under the theorem's local assumptions, unit steps are eventually accepted for $\sigma < 1/2$. There is no unconditional global-convergence claim.

9.9 Exercises

Exercise 9.1 (Operator-norm estimates). Starting from the induced norm definition, prove

$$\|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\|_2 \|\mathbf{x}\|, \quad \|\mathbf{A}\mathbf{B}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_2.$$

Identify where both inequalities enter the proof of [Theorem 9.1](#).

Exercise 9.2 (Recognizing descent directions). Let

$$f(x_1, x_2) := x_1^2 + 4x_2^2, \quad \mathbf{x}^{(0)} := (1, 1)^\top.$$

For each of $\mathbf{d}_1 = (-1, 0)^\top$, $\mathbf{d}_2 = (1, -1)^\top$, and $\mathbf{d}_3 = (-2, -1)^\top$, determine whether it is a strict descent direction at $\mathbf{x}^{(0)}$. For every descending choice, find an interval of positive step lengths that actually decreases f .

Exercise 9.3 (Newton on a positive-definite quadratic). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be symmetric positive definite and

$$f(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{c}^\top \mathbf{x}.$$

Starting from arbitrary $\mathbf{x}^{(0)} \in \mathbb{R}^n$, determine the first full Newton iterate and hence the number of updates required to reach the unique global minimizer.

Exercise 9.4 (A positive Hessian does not guarantee full-step decrease). For $f(x) := x + \frac{1}{2}x^2 + 6x^4$, compute the Newton direction at $x^{(0)} = 0$, verify that it descends, and determine all $\alpha > 0$ for which $f(x^{(0)} + \alpha d^{(0)}) < f(x^{(0)})$.

Exercise 9.5 (A singular limiting Hessian). Let

$$f(x_1, x_2) := (x_1^2 + x_2^2)^2, \quad \mathbf{x}^{(0)} := (a, a)^\top, \quad a \neq 0.$$

Show that every Newton iterate is well defined and

$$\mathbf{x}^{(k)} = \left(\frac{2}{3}\right)^k (a, a)^\top.$$

Prove that $\mathbf{0}$ is the unique strict global minimizer, compute its Hessian, and explain why [Theorem 9.1](#) does not apply.

Exercise 9.6 (A smooth Newton two-cycle). Let $f(x) := x^4/4 - x^2 + 2x$ on $\mathcal{D} = \mathbb{R}$.

- (a) Derive Newton's map, state where it is defined, and verify the cycle $0, 1, 0, 1, \dots$
- (b) Show that the only stationary point x^* lies in $(-2, -1)$ and is the unique global minimizer. *Hint:* analyze the turning points of f' ; no cubic formula is needed.
- (c) Check [Theorem 9.1](#) at x^* . Why does the cycle not contradict it?

Exercise 9.7 (One shifted-Newton direction). Suppose at an iterate

$$\mathbf{g}^{(k)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{H}^{(k)} = \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix}.$$

- (a) Compute the unmodified Newton direction and test whether it descends.
- (b) Find the smallest integer $\tau \geq 0$ for which $\mathbf{H}^{(k)} + \tau I$ is positive definite.
- (c) Solve the shifted system using that integer and verify descent directly.
- (d) Compare the normalized shifted direction for $\tau = 10^3$ with the normalized negative gradient.

Exercise 9.8 (Exact line search). Let $\mathbf{x}^{(0)} = (0, 0)^\top$, $\mathbf{d}^{(0)} = (1, -1)^\top$, and

$$f(x_1, x_2) := (x_1 - 2)^2 + 2(x_2 + 1)^2 + x_1 x_2.$$

Find the exact-line-search step over $\alpha \geq 0$, and verify that the argmin is attained.

Exercise 9.9 (Armijo trials). Let $f(x) := x + \frac{1}{2}x^2 + 2x^4$, $x^{(0)} = 0$, and let $d^{(0)}$ be the Newton direction. With $\bar{\alpha} = 1$, $\beta = 1/2$, and $\sigma = 10^{-4}$, list the Armijo trials until the first accepted step. Compare its objective value with that of the full Newton trial.